Lab 11 Activity

PSYC 6802: Intro to Psychological Statistics

We will work with the same dataset that we used in the main lab. You can read in the data using the following code:

```
library(tidyverse)
library(haven)
library(ggeffects)
library(lavaan)
dat <- read_dta("Lab_11_attend.dta")
```

1

The variables `frosh` and `soph` are dummy variables for the student being a freshman or sophomore (0 on both variables indicates 'other'). Run the code below to reconstruct a factor variable (with "Other" as the default category) from the dummy variables.

```
dat <- dat %>%
  mutate(grade = case_when(frosh == 1 ~ "Freshman",
                           soph == 1 ~ "Sophomore",
                           .default = "Other"),
         grade = fct_relevel(grade, "Other"))
```

2

Fit a model predicting final exam scores from (the centered) prior GPA with academic year as a moderator (no need to center the categorical moderator variables). Are there statistically significant moderation effects?

3

Construct plots to illustrate the moderation effect (use separate lines for different grade levels). How does this plot help you interpret your results from step 1?

4

Using this same data set, conduct a mediation analysis predicting final exam scores from ACT scores with `hwrte` (rate of completing homework) as a mediator (you may drop the 6 missing values of `hwrte` for this step). What do you conclude?

5

Repeat #3 using a method that using a better method of missing data handling than listwise deletion.